FRM PART I – MOCK II ANSWERS

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By AnalystPrep

Reference Table: Let Z be a standard normal random variable.

z	P(Z<z)	z	P(Z<z)	z	P(Z<z)	z	P(Z<z)	z	P(Z<z)	z	P(Z<z)
-3	0.0013	-2.50	0.0062	-2.00	0.0228	-1.50	0.0668	-1.00	0.1587	-0.50	0.3085
-2.99	0.0014	-2.49	0.0064	-1.99	0.0233	-1.49	0.0681	-0.99	0.1611	-0.49	0.3121
-2.98	0.0014	-2.48	0.0066	-1.98	0.0239	-1.48	0.0694	-0.98	0.1635	-0.48	0.3156
-2.97	0.0015	-2.47	0.0068	-1.97	0.0244	-1.47	0.0708	-0.97	0.1660	-0.47	0.3192
-2.96	0.0015	-2.46	0.0069	-1.96	0.0250	-1.46	0.0721	-0.96	0.1685	-0.46	0.3228
-2.95	0.0016	-2.45	0.0071	-1.95	0.0256	-1.45	0.0735	-0.95	0.1711	-0.45	0.3264
-2.94	0.0016	-2.44	0.0073	-1.94	0.0262	-1.44	0.0749	-0.94	0.1736	-0.44	0.3300
-2.93	0.0017	-2.43	0.0075	-1.93	0.0268	-1.43	0.0764	-0.93	0.1762	-0.43	0.3336
-2.92	0.0018	-2.42	0.0078	-1.92	0.0274	-1.42	0.0778	-0.92	0.1788	-0.42	0.3372
-2.91	0.0018	-2.41	0.0080	-1.91	0.0281	-1.41	0.0793	-0.91	0.1814	-0.41	0.3409
-2.9	0.0019	-2.40	0.0082	-1.90	0.0287	-1.40	0.0808	-0.90	0.1841	-0.40	0.3446
-2.89	0.0019	-2.39	0.0084	-1.89	0.0294	-1.39	0.0823	-0.89	0.1867	-0.39	0.3483
-2.88	0.0020	-2.38	0.0087	-1.88	0.0301	-1.38	0.0838	-0.88	0.1894	-0.38	0.3520
-2.87	0.0021	-2.37	0.0089	-1.87	0.0307	-1.37	0.0853	-0.87	0.1922	-0.37	0.3557
-2.86	0.0021	-2.36	0.0091	-1.86	0.0314	-1.36	0.0869	-0.86	0.1949	-0.36	0.3594
-2.85	0.0022	-2.35	0.0094	-1.85	0.0322	-1.35	0.0885	-0.85	0.1977	-0.35	0.3632
-2.84	0.0023	-2.34	0.0096	-1.84	0.0329	-1.34	0.0901	-0.84	0.2005	-0.34	0.3669
-2.83	0.0023	-2.33	0.0099	-1.83	0.0336	-1.33	0.0918	-0.83	0.2033	-0.33	0.3707
-2.82	0.0024	-2.32	0.0102	-1.82	0.0344	-1.32	0.0934	-0.82	0.2061	-0.32	0.3745
-2.81	0.0025	-2.31	0.0104	-1.81	0.0351	-1.31	0.0951	-0.81	0.2090	-0.31	0.3783
-2.8	0.0026	-2.30	0.0107	-1.80	0.0359	-1.30	0.0968	-0.80	0.2119	-0.30	0.3821
-2.79	0.0026	-2.29	0.0110	-1.79	0.0367	-1.29	0.0985	-0.79	0.2148	-0.29	0.3859
-2.78	0.0027	-2.28	0.0113	-1.78	0.0375	-1.28	0.1003	-0.78	0.2177	-0.28	0.3897
-2.77	0.0028	-2.27	0.0116	-1.77	0.0384	-1.27	0.1020	-0.77	0.2206	-0.27	0.3936
-2.76	0.0029	-2.26	0.0119	-1.76	0.0392	-1.26	0.1038	-0.76	0.2236	-0.26	0.3974
-2.75	0.0030	-2.25	0.0122	-1.75	0.0401	-1.25	0.1056	-0.75	0.2266	-0.25	0.4013
-2.74	0.0031	-2.24	0.0125	-1.74	0.0409	-1.24	0.1075	-0.74	0.2296	-0.24	0.4052
-2.73	0.0032	-2.23	0.0129	-1.73	0.0418	-1.23	0.1093	-0.73	0.2327	-0.23	0.4090
-2.72	0.0033	-2.22	0.0132	-1.72	0.0427	-1.22	0.1112	-0.72	0.2358	-0.22	0.4129
-2.71	0.0034	-2.21	0.0136	-1.71	0.0436	-1.21	0.1131	-0.71	0.2389	-0.21	0.4168
-2.7	0.0035	-2.20	0.0139	-1.70	0.0446	-1.20	0.1151	-0.70	0.2420	-0.20	0.4207
-2.69	0.0036	-2.19	0.0143	-1.69	0.0455	-1.19	0.1170	-0.69	0.2451	-0.19	0.4247
-2.68	0.0037	-2.18	0.0146	-1.68	0.0465	-1.18	0.1190	-0.68	0.2483	-0.18	0.4286
-2.67	0.0038	-2.17	0.0150	-1.67	0.0475	-1.17	0.1210	-0.67	0.2514	-0.17	0.4325
-2.66	0.0039	-2.16	0.0154	-1.66	0.0485	-1.16	0.1230	-0.66	0.2546	-0.16	0.4364
-2.65	0.0040	-2.15	0.0158	-1.65	0.0495	-1.15	0.1251	-0.65	0.2578	-0.15	0.4404
-2.64	0.0041	-2.14	0.0162	-1.64	0.0505	-1.14	0.1271	-0.64	0.2611	-0.14	0.4443
-2.63	0.0043	-2.13	0.0166	-1.63	0.0516	-1.13	0.1292	-0.63	0.2643	-0.13	0.4483
-2.62	0.0044	-2.12	0.0170	-1.62	0.0526	-1.12	0.1314	-0.62	0.2676	-0.12	0.4522
-2.61	0.0045	-2.11	0.0174	-1.61	0.0537	-1.11	0.1335	-0.61	0.2709	-0.11	0.4562
-2.6	0.0047	-2.10	0.0179	-1.60	0.0548	-1.10	0.1357	-0.60	0.2743	-0.10	0.4602
-2.59	0.0048	-2.09	0.0183	-1.59	0.0559	-1.09	0.1379	-0.59	0.2776	-0.09	0.4641
-2.58	0.0049	-2.08	0.0188	-1.58	0.0571	-1.08	0.1401	-0.58	0.2810	-0.08	0.4681
-2.57	0.0051	-2.07	0.0192	-1.57	0.0582	-1.07	0.1423	-0.57	0.2843	-0.07	0.4721
-2.56	0.0052	-2.06	0.0197	-1.56	0.0594	-1.06	0.1446	-0.56	0.2877	-0.06	0.4761
-2.55	0.0054	-2.05	0.0202	-1.55	0.0606	-1.05	0.1469	-0.55	0.2912	-0.05	0.4801
-2.54	0.0055	-2.04	0.0207	-1.54	0.0618	-1.04	0.1492	-0.54	0.2946	-0.04	0.4840
-2.53	0.0057	-2.03	0.0212	-1.53	0.0630	-1.03	0.1515	-0.53	0.2981	-0.03	0.4880
-2.52	0.0059	-2.02	0.0217	-1.52	0.0643	-1.02	0.1539	-0.52	0.3015	-0.02	0.4920
-2.51	0.0060	-2.01	0.0222	-1.51	0.0655	-1.01	0.1562	-0.51	0.3050	-0.01	0.4960

Reference Table: t-table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

FRM Part I – Mock Exam II –Candidate Answer Sheet

1.		26.		51.		76.	
2.		27.		52.		77.	
3.		28.		53.		78.	
4.		29.		54.		79.	
5.		30.		55.		80.	
6.		31.		56.		81.	
7.		32.		57.		82.	
8.		33.		58.		83.	
9.		34.		59.		84.	
10.		35.		60.		85.	
11.		36.		61.		86.	
12.		37.		62.		87.	
13.		38.		63.		88.	
14.		39.		64.		89.	
15.		40.		65.		90.	
16.		41.		66.		91.	
17.		42.		67.		92.	
18.		43.		68.		93.	
19.		44.		69.		94.	
20.		45.		70.		95.	
21.		46.		71.		96.	
22.		47.		72.		97.	
23.		48.		73.		98.	
24.		49.		74.		99.	
25.		50.		75.		100.	

Foundations of Risk Management

1. Next week, Global Tech has declared its intention to bring to the market a 15-year senior bond issue at par with a coupon rate of 18%, offering a spread of 800 basis points over the corresponding 15-year Treasury issue. An investor is keen to enter into a total return swap that matures in one year with the senior bonds that are about to be issued as the reference obligation. Under the terms of the contract, payments will be exchanged semiannually, where the total return receiver will pay the six-month Treasury rate plus 350 basis points.

What is the 15-year Treasury rate at the time the bonds are issued?

A. 8%
B. 2%
C. 10%
D. 4.5

2. A portfolio has a return of 15%, and its standard deviation is 25%. If the return on the market is 8% with a standard deviation of 50% and the risk-free rate is 5%, what are the Sharpe and Treynor ratios for the portfolio?

A. Sharpe ratio: 0.40; Treynor ratio: 0.13
B. Sharpe ratio: 0.28; Treynor ratio: 0.09
C. Sharpe ratio: 0.40; Treynor ratio: 0.20
D. Sharpe ratio: 0.28; Treynor ratio: 0.09

3. You have been given the following information regarding a portfolio:

Return on the portfolio	12%
Return on the market	8%
Return on benchmark	9%
Beta of the portfolio	1.1
Tracking error of the portfolio	10%
Standard deviation of the portfolio	15%

Which of the following is closest to the information ratio of the portfolio?

A. 0.3
B. 0.2
C. 0.4
D. 0.5

4. The collapse of Enron was most likely a result of:

A. Unauthorized trading activity
B. Unanticipated market moves
C. Dubious accounting practices
D. All of the above

5. Which of the following cases most likely involved the mismanagement of hedging strategies?

A. Enron
B. Volkswagen
C. Metallgesellschaft
D. Barings Bank

6. The 2007-2008 financial crisis was primarily a result of two major risks. These are:

A. Credit and liquidity risk
B. Credit and market risk
C. Liquidity and operational risk
D. Market and operational risk

7. A bank only uses VaR to measure the risk in its trading book. The bank has a one-day VaR of $25 million at a confidence level of 99%. During the last 200 trading days, this VaR was exceeded only twice, with total losses of $500 million and $200 million being incurred. According to the VaR measure on its own, the bank has been successful in managing its risk but, in reality, it has lost $700 million.

In this scenario, the failure of risk management can *most likely* be attributed to:

A. Failure to use appropriate risk metrics
B. Mismeasurement of known risk
C. Failure to take known risks into account
D. Failure in communicating risk to top management

8. Which of these correctly describes the numerator of the Sortino measure?

A. Difference between portfolio return and market return
B. Difference between portfolio return and minimum acceptable return
C. Difference between portfolio return and risk-free rate
D. Difference between portfolio return and benchmark return

9. Which of the following parts of GARP's Code of conduct requires members to comply with all applicable laws, rules, and regulations (including the Code itself) governing the GARP Members' professional activities?

A. Fundamental responsibilities
B. Professional integrity and ethical conduct
C. Conflict of interest
D. General accepted principles

10. Jane Johnson, FRM, works as a full-time risk manager for the Washington Investment Banking Group. Recently, one of Johnson's long-time friends asked her to help with preparations for an initial public offering (IPO) for a promising tech company based in Silicon Valley. This would be a part-time role and would entail establishing earnings projections for the tech company. Johnson should most likely:

A. Accept the work as long as the Washington Investment Banking Group agrees to all the terms of engagement
B. Accept the work as long as she obtains written consent from the Washington Investment Banking Group and does it in her free time
C. Accept the role as long as, in her own assessment, it does not interfere with her duties to the Washington Investment Banking Group
D. Not accept the work as it violates the Code of Conduct by creating a conflict of interest

11. Peter Drury, FRM, serves as the chief risk officer at Capital bank. He recently finalized a comprehensive risk assessment on a series of investments in credit default swaps. According to his estimates, the portfolio had a 1% chance of losing USD 500 million or more over one year, a big enough loss to trigger insolvency or attract takeover bids. Based on Drury's assessment, Capital Bank's CEO authorized a large investment in additional swaps. At the end of the first year, the portfolio had lost USD 650 million. The bank was immediately closed down by the regulator, pending a comprehensive audit and review of the bank's investment policies.

Which of the following statements is correct?

A. The outcome demonstrates risk management failure because the CRO failed to foresee a move by the regulator, neither did he attempt to stop the shutdown
B. The outcome demonstrates an outright risk management failure because the occurrence of a supposedly extreme event means the probability of such an outcome was poorly estimated
C. The outcome demonstrates risk management failure because the CRO did not eliminate the chance of a financial loss
D. On the basis if the information provided, one cannot conclusively determine whether this was a risk management failure

12. A risk manager at an investment consultancy firm is projecting a return of 14% on Portfolio Y. The market risk premium is 15%, the risk-free rate on interest is 5%, and the volatility of the market portfolio is 18%. Portfolio Y has a beta of 0.6. According to the Capital Asset Pricing Model, which of the following statements is true?

A. The expected return of portfolio Y is more than the expected return of the market portfolio
B. The return of the portfolio has more volatility than the market portfolio
C. The expected return of portfolio Y is less than the expected return of the market portfolio
D. The expected return of portfolio Y is equal to the expected return of the market portfolio

13. Catherine Austin works at the arbitrage department of an investment bank. She finds that an asset is trading at USD 2,200.The price of a one-year futures contract on the asset is USD2,244, and the price of a two-year futures contract is USD 2,295.50. Assume that the asset exhibits no cash flows in the first two years. If the term structure of interest rates is flat at 2%, which of the following would be an appropriate arbitrage strategy?

A. Short the 2-year futures, borrow USD 2,000 at 2% and buy the underlying asset
B. Buy the 2-year futures and short the underlying asset
C. Short the 2-year futures, borrow USD 2,000 at 2% and short the underlying asset
D. Buy the 2-year futures, borrow USD 2,000 at 2% and short the underlying asset

14. A wealth management firm is contemplating motivational measures in an attempt to align the interests of managers and shareholders. To this end, the board has plans to grant managers securities that are tied to the firm's stock. Even as it ponders such a move, the board is wary of managerial incentives that could negatively affect risk management at the firm. Which of the following securities provides the best example of rewards likely to reduce managerial incentives to manage risk?

A. A long position in the firm's stock
B. A deep out-of-the-money call option on the firm's stock
C. At-the-money call option on the firm's stock
D. A deep in-the-money call option on the firm's stock

15. The failure of Long Term Capital Management almost brought the financial markets to a standstill and has since become a marker buoy to show just how far certain risk management safeguards can go to alleviate the risk of failure. Which of the following statements about LTCM's failure is incorrect?

A. LTCM relied too much on theoretical market risk models and too little on stress testing, gap risk, and liquidity risk
B. LTCM scantily disclosed its positions and exposures, and counterparties had few ways to find out LTCM's exposure to other counterparties
C. In most of its repo and swap transactions' LTCM did not pay an initial margin as would other non-bank counterparties
D. There was no moral hazard at play in the run-up to LTCM's failure

16. A portfolio manager returns 20% with a volatility of 30%. The benchmark returns 15% with a risk of 22%. The correlation between the two is 0.80, while the risk-free rate is 5%. Which of the following statements is correct?

A. The portfolio has a lower SR than the benchmark
B. The IR of the portfolio is negative
C. IR = 0.35
D. IR = 0.28

17. Patricia Rice, FRM, is evaluating the risk management of Bright Technologies. She is asked to categorize the following events into various types of risk. Identify each numbered event as a credit risk, market risk, business risk, operational risk, and legal risk.

1. A delayed launch of the latest model of an internet connectivity gadget leads to an inventory pileup of the older product and a USD 100 million write-off of excess inventory.

2. Insufficient training leads to misuse of the check clearing system

3. Credit spreads widen following bankruptcies

4. Swap seller does not have the resources required to honor a contract

5. Credit swaps cannot be netted because they originated in Canada and Sydney

A. 1: operational risk; 2: business risk; 3: credit risk; 4: market risk; 5: credit risk
B. 1: business risk; 2: operational risk; 3: credit risk; 4: market risk; 5: credit risk
C. 1: business risk; 2: operational risk; 3: market risk; 4: credit risk; 5: legal risk
D. 1: operational risk; 2: business risk; 3: credit risk; 4: market risk; 5: legal risk

18. Which of the following risk management responsibilities rest with the board of directors of a bank?

I. Approval of the bank's risk management policies and procedures
II. Maintaining oversight of all risk management activities
III. Evaluating the performance of risk management activities
IV. Ensuring compliance with minimum or best practice standards

A. I and II only
B. I and III only
C. II and IV only
D. I, II, III, and IV

19. Mathew Jelkins, FRM, was recently found guilty of violating the principle of professional integrity and ethical conduct as outlined in the GARP Code of Conduct. Which of the following is a potential consequence? Jelkins will be:

A. Required to participate in at least one ethical training workshop
B. Stripped of the GARP member's right to use the FRM designation
C. Stripped of the GARP member's right to work in the risk management profession
D. Required to pay a fine whose amount will be determined by the GARP's penalty committee

20. You have been given the following data for a managed portfolio:

Beta	1.55
Alpha	2.11%
Average return	12.5%
Risk-free rate	3%

Basing your calculations on Jensen's measure of portfolio performance, the return on the market portfolio is closest to:

A. 1.77%
B. 7.77%
C. 8.36%
D. 5.36%

Quantitative Analysis

21. Which of the following statements is false?

A. The result of a single toss of a fair coin is a mutually exclusive event
B. The results of two coin tosses of a fair coin are independent events
C. The results of drawing two cards from a deck of cards are independent events
D. The results of two rolls of a fair die are independent events

22. You have been given the following incomplete probability matrix about the return (in foreign currency terms) on an investment made in a foreign country as compared to the benchmark and the expected change in the exchange rate of the local currency against the foreign currency:

	Outperform	**Underperform**
Appreciation	10%	15%
No change		20%
Depreciation	25%	
Total		55%

What is the unconditional probability that the investment will outperform the market?

A. 45%
B. 40%
C. 55%
D. 50%

23. A random variable Y has an equal probability of having a value of 0, 1, 2 or -1. Given that X = 3Y, what is the covariance of Y and X?

A. 3.25
B. 3.00
C. 3.50
D. 3.75

24. All of the following statements are true, except:

A. If the return distributions of two investments have the same mean and standard deviation, the one which has more a negatively skewed distribution will be considered to be riskier
B. Distributions with positive excess kurtosis are known as leptokurtic
C. If two investments have the same mean, standard deviation and skewness, then the one with the lower kurtosis will be considered riskier
D. The normal distribution has a kurtosis of 3

25. Which of these probability distributions is most likely to be used to model stock prices?

A. Normal
B. Lognormal
C. Binomial
D. Poisson

26. In 2016, DollarCom issued both BBB corporate bonds and common stocks. The following probability matrix summarizes the performance of the stock according to the ratings of the bonds:

		Stock Outperform	**Stock Underperform**
	Upgrade to BBB+	15%	5%
Bonds	**No change**	10%	20%
	Downgrade to BBB-	20%	30%

What is the probability that the credit ratings of the bonds are upgraded, given that the stock has underperformed?

A. 20%
B. 25%
C. 9%
D. 5%

27. You have been given the following probabilities:

P [A | B] = 25%

P [B | A] = 50%

P [A] = 30%

What is P[B]?

A. 60%
B. 15%
C. 75%
D. 42%

28. An analyst gathers monthly data about the returns of a stock for the past five years. If the mean monthly return is 6% and the standard deviation of the series of returns is 1.8%, then what is the standard deviation of the mean over the period?

A. 0.45%
B. 0.23%
C. 0.52%
D. 1.39%

29. An insurance policy reimburses dental expenses, X, up to a maximum benefit of 250. The probability density function for X is as follows:

$$f_X(x) = \begin{cases} ke^{-0.004x}, & x \geq 0 \\ 0, & elsewhere \end{cases}$$

Where k is a constant.

Calculate the median amount of the benefit.

A. 160
B. 172
C. 173
D. 250

30. A zero-coupon bond with a notional value of $100 and price x has the following probability density function:

$$f(x) = \frac{x}{5000} \; for \; 0 \leq x \leq 100$$

Determine the probability that the price of the bond is between $80 and $90 inclusively.

A. 0.0017
B. 0.81
C. 0.17
D. 0.64

31. For a variable v, the value of α is 2, and the probability that v > 100 is 0.1. What is the probability that v is greater than 200?

A. 0.025
B. 0.050
C. 0.105
D. 0.225

32. You have been provided the following set of values for an independent variable A and a dependent variable B:

A	**B**
1	5
2	8
3	12
4	15

What are the slope and the intercept of the regression line?

A. The slope is 3.4, and the intercept is 1.5
B. The slope is 2.6, and the intercept is 2.2
C. The slope is 3.1, and the intercept is 1.3
D. The slope is 2.8, and the intercept is 1.8

33. An ordinary least squares regression produces the following:

$$\sum(\boldsymbol{Y_i} - \overline{\boldsymbol{Y}})^2 = \mathbf{58}$$

$$\sum(\boldsymbol{Y_i} - \widehat{\boldsymbol{Y}})^2 = \mathbf{10}$$

$$\sum(\widehat{\boldsymbol{Y}} - \overline{\boldsymbol{Y}})^2 = \mathbf{48}$$

What is the coefficient of determination for the regression?

A. 0.792
B. 0.208
C. 0.828
D. 0.654

34. The result of a two-variable regression analysis produces the following data:

	Coefficient	
Constant	2.1	
Slope	1.2	

	Degrees of freedom	Sum of squares
Regression	1	35.4
Residual error	8	14.6

What is the correlation coefficient of the two variables?

A. 0.841
B. 0.708
C. 0.588
D. 0.767

35. An analyst performs a regression of a stock's return on the overall market return using a sample of 2 years of monthly returns. He estimates the slope coefficient to be 0.55 with a standard deviation of 0.26. The analyst then performs a hypothesis test at a 5% level of significance to determine if the slope coefficient is significantly different from zero. Which of the following correctly describes the test statistic and result of this test?

A. The test statistic is 2.12, and the null hypothesis is rejected to conclude that the slope coefficient is different from zero
B. The test is 2.12, and the null hypothesis cannot be rejected to conclude that the slope coefficient is different from zero
C. The test is 0.47, and the null hypothesis cannot be rejected to conclude that the slope coefficient is different from zero
D. The test is 0.47, and the null hypothesis is rejected to conclude that the slope coefficient is different from zero

36. If the variance of the residuals is not constant across all observations in a sample, the regression is said to be:

A. Leptokurtic
B. Multicollinear
C. Platykurtic
D. Heteroskedastic

37. Which of the following statements is false?

A. Multicollinearity occurs when two or more of the independent variables in a multiple regression are highly correlated with each other
B. Multicollinearity distorts the standard error of the regression and the coefficient of standard errors
C. If there exists imperfect multicollinearity, then it is not possible to find the ordinary least squares (OLS) estimators necessary for the regression results
D. As a result of multicollinearity, there is a greater probability that it will be incorrectly concluded that a variable is not statistically significant

38. An analyst runs a multiple regression of a variable Y on four independent variables for 50 observations. If the sum of squared errors for the regression is 125 and the total sum of squares is 450, then what is the adjusted R^2 of the regression?

A. 0.722
B. 0.786
C. 0.697
D. 0.643

39. If the leading and lagged values of the mean, variance and covariance of a time series do not change with time, the time series is said to be:

A. Autocorellated
B. Autoregressive
C. Covariance stationary
D. Autocovariant

40. Which of the following statements is (are) false?

I. An advantage of bootstrapping is that it allows to make inferences without making strong assumptions about the distribution

II. Another advantage of bootstrap is that it is not affected by outliers in the data

III. Bootstrapping will be ineffective if the data are not independent of one another

A. I only
B. II only
C. II & III only
D. All of the statements are false

Financial Markets and Products

41. An unwanted result of deposit insurance is that it can cause banks to engage in risky behavior in the knowledge that depositors are well insured by the *Federal Deposit Insurance Corporation* (*FDIC*). This is an example of:

A. Adverse selection
B. Moral hazard
C. Asymmetric information
D. Systemic risk

42. An insurance contract that lasts for a specified period and pays a lump sum either when the policyholder dies or at the end of the period, whichever comes first, is known as:

A. Term life insurance
B. Whole life insurance
C. Endowment life insurance
D. Universal life

43. An insurance company's financial statements contain the following information:

	(In million USD)
Premiums	120
Claims	65
Operating expenses	20
Investment income	5

The combined ratio for the insurance company is closest to:

A. 54%
B. 75%
C. 68%
D. 71%

44. An investor in a hedge fund requires a net return after fees of 18%. If the hedge fund charges fees of 1 plus 25%, how much will the hedge fund have to earn for the investor to get his desired return?

A. 22.30%
B. 25.00%
C. 25.70%
D. 24.00%

45. Which of the following investment strategies does an equity market neutral fund follow?

A. Long-short equity
B. Dedicated short
C. Merger arbitrage
D. Convertible arbitrage

46. An investor buys a European put option on the shares of Yullu Corp. for a strike price of $25. Suppose the price of the share price falls to $20 at the time of the maturity of the option. If the cost of the put option was $2, then what is the net profit for the investor?

A. $20
B. $0
C. $3
D. $5

47. You have been given the following spot rates table:

Maturity	Spot Rate
0.5 year	4.0%
1 year	4.3%
1.5 years	4.9%
2 years	5.5%

If a 2-year bond with a principal of $1000 pays a semiannual coupon of 8% per year, then which of the following is closest to the theoretical price of the bond?

A. $1,046
B. $998
C. $1,002
D. $1,197

48. Two investors – A and B – invest $1000 each in two different financial instruments. Investor A receives a return of 5% with semiannual compounding while Investor B receives a return of 5% with quarterly compounding. After two years, what will be the terminal values of the two investments?

	Investor A	Investor B
A.	$1,103.8	$1,104.4
B.	$1,104.4	$1,103.8
C.	$1,215.5	$1,477.5
D.	$1,477.5	$1,215.5

49. What is the duration of a 1.5-year 6% semiannual coupon bond with a face value of $100, if the yield on the bond is 10% per annum with continuous compounding?

A. 1.3918
B. 1.4009
C. 1.4553
D. 1.4949

50. An investor can immunize their portfolio against large parallel shifts in the zero curve by:

A. Choosing a portfolio of assets and liabilities with a net duration of 1, and net convexity of 0.
B. Choosing a portfolio of assets and liabilities with a net duration of 0, and net convexity of 1.
C. Choosing a portfolio of assets and liabilities with a net duration of 0, and net convexity of 0.
D. Choosing a portfolio of assets and liabilities with a net duration of 1, and net convexity of 1.

51. An investor short sells 1,000 shares of XYZ Corporation in June when the price is $50 per share. In September, a dividend of $1.5 per share is paid by the company. The investor closes out the position in October when the price of the share is $45. Assuming there are no borrowing costs involved, what is the profit or loss for the investor?

A. $3,500
B. $6,500
C. $5,000
D. $1,500

52. An asset that provides no income has a storage cost of $3 per unit, paid at the end of the year. If the spot price of the asset is $220 and the risk-free rate is 5%, the futures price for the asset is closest to:

A. $228.80
B. $238.11
C. $231.28
D. $234.28

53. When the futures price is above the expected future spot price, the situation is known as

A. Backwardation
B. Contango
C. Cost of carry
D. None of the above

54. You have been given the following information regarding a stock index:

Current index level	$1,120
Risk-free rate	6% per annum
Dividend yield	2% per annum

What is the futures price for a six-month contract on the index?

A. $1,142.63
B. $1,165.70
C. $1,154.11
D. $1,137.60

55. You have been given the following details for two bonds:

	US Treasury Bond	Corporate Bond
Principal	$100	$100
Coupon	5%	6%

If coupons are paid on March 5 and September 5 of each year for both bonds, what is the interest accrued on both bonds between the period of March 5, 2017, and May 27, 2017?

A. $1.128 on the Treasury Bond and $1.353 on the Corporate Bond.
B. $1.128 on the Treasury Bond and $1.367 on the Corporate Bond.
C. $1.139 on the Treasury Bond and $1.353 on the Corporate Bond.
D. $1.139 on the Treasury Bond and $1.367 on the Corporate Bond.

56. The price of a Treasury bond, which has a coupon of 10%, is quoted as 103-05. If the current date is March 28, 2018, and the bond will mature on October 25, 2030, the cash price of the bond is closest to:

A. $107.43
B. $111.66
C. $103.46
D. $112.24

57. Which of the following bond is the cheapest to deliver if the Treasury bond futures price is 100-08?

	Price	Conversion Factor
A.	122-04	1.2001
B.	134-08	1.3190
C.	118-20	1.1568
D.	141-24	1.3987

58. An increase in one of the following factors is likely to cause the price of an American call option to decrease. Which one?

A. Current stock price
B. Strike price
C. Volatility
D. Risk-free rate

59. Which of the following correctly describes a protective put?

A. Long position in a put option and a short position in the underlying shares
B. Long position in a put option and a long position in the underlying shares
C. Short position in a put option and a long position in the underlying shares
D. Short position in a put option and short position in the underlying shares

60. A European call option on a non-dividend paying stock has the following details:

Current stock price	$21
Strike price	$20
Time to maturity	6 months
Risk-free interest rate	10%

What is the lower bound for the option price?

A. $1.32
B. $0
C. $1.98
D. $2.90

61. You have been given the following information:

Price of European call option on stock X	$1.5
Maturity of call option on stock X	6 months
Strike price of call option on stock X	$42
Current price of a share of stock X	$38
Expected dividend on stock X (to be paid two months from now)	$0.75
Risk-free interest rate	6%

What is the price of a European put option that expires in 6 months and has a strike price of $42?

A. $5.00
B. $4.50
C. $3.76
D. $4.00

62. An increase in which of the following factors is likely to cause the price of a European put option to increase?

A. Current stock price, dividends and volatility
B. Strike price, dividends and volatility
C. Dividends, risk-free rate and volatility
D. Risk-free rate and volatility

63. A strategy where a long position in a call option is combined with a long position in a put option, such that both options have the same strike price and expiration date, is known as

A. Strangle
B. Bear spread
C. Straddle
D. Bearish option

64. An investor wants to implement a strangle using put and call options on the shares of UUKL. Which of the following pairs of options is he most likely to choose?

A. Buy a call option with a strike price of $30 expiring in 6 months and a buy a put option with a strike price of $30 expiring in 6 months
B. Buy a call option with a strike price of $30 expiring in 6 months and a buy a put option with a strike price of $30 expiring in 3 months
C. Buy a call option with a strike price of $30 expiring in 6 months and a buy a put option with a strike price of $25 expiring in 6 months
D. Buy a call option with a strike price of $30 expiring in 6 months and a buy a put option with a strike price of $25 expiring in 3 months

65. An investor who buys a put option with a strike price of $25 and sells a put option with a strike price of $20 is most likely following which of these strategies:

A. Bull spread
B. Box spread
C. Butterfly spread
D. Bear spread

66. Suzy Wong, a finance student at the University of Tokyo, regularly invests her extra income in stocks and derivatives. She owns stocks of Safe Home Cleaning Inc., which is currently trading at $29. She believes the stock will trade below $29 if new regulations on cleaning companies are introduced next month. If she is interested in entering in an option position that gives her the right to sell her stocks at $29 if the price of the stock goes below $29, then suggest the most appropriate option position for Wong.

A. Long call option with the strike price of $29
B. Short call option with the strike price of $29
C. Long put option with the strike price of $29
D. Short put option with the strike price of $29

67. Which of the following statements about over-the-counter (OTC) derivatives is false?

A. OTC derivatives have flexible and negotiable terms
B. OTC derivatives have very good liquidity
C. The credit risk for OTC derivatives is bilateral
D. OTC derivatives have negotiable and non-standard maturities

68. Which of these statements about exchange-traded derivatives is false?

A. Exchange-traded derivatives have standardized contract terms
B. Exchange-traded derivatives have low liquidity risk
C. Exchange-traded derivatives have high credit risk
D. Exchange-traded derivatives have standard maturities

69. Which of these is most likely to be a disadvantage of having a central counterparty (CCP)?

A. Liquidity risk
B. Loss mutualisation
C. Lack of transparency
D. Adverse selection

70. The key risk for a Central Counterparty (CCP) is that of default of a clearing member. Such an event can also have certain knock-on effects. These knock-on effects can include all of the following, EXCEPT:

A. Failed auctions
B. Default of other clearing members
C. Resignations
D. Wrong-way risk

Valuation and Risk Models

71. Frank Ort is a junior trader at the options trading desk of an investment bank. He is reviewing a recent analysis of Tempo Construction LLC. Stocks of Tempo are currently trading at $47. According to the bank's analysts, the price of the stock will either go up or down by $8 in one year depending on the revenues that the company will generate. Since Ort is quite pessimistic on the future development of the company, he is thinking about opening a short position by buying put options on Tempo's stocks with a strike price of $45. The risk-free rate is assumed to be 2.1% per year.

What is the no-arbitrage price of the option that Ort is looking to buy?

A. $2.55
B. $2.57
C. $2.77
D. $2.97

72. An analyst at an options trading desk is valuing a 3-month European call option on the broad market value-oriented index. The index is currently trading at $14 and has a continuously compounded dividend rate of 1%. The analyst expects a price volatility of 12% per quarter. The risk-free rate is 5% per year, and the option has a strike price of $14.

What is the price of the European call option?

A. $0.9022
B. $0.9045
C. $0.9207
D. $0.9232

73. Cytadela Bank from Poznan, Poland, needs to measure the yield of a coupon bond portfolio. One of the bonds in the portfolio has the maturity exactly equal to the term of a key rate, and the price of that bond is exactly par. How do the bond's yield and the key rate compare?

A. They are identical
B. The bond's yield is higher than the key rate
C. The bond's yield is lower than the key rate
D. The bond's yield has no discernible relationship with the key rate

74. You have been provided the following information:

Daily volatility estimate	2%
Decay factor	0.75
Most recent market return	3%

Using the Exponentially Weighted Moving Average Method (EWMA), what is the new estimate for volatility?

A. 2.101%
B. 2.783%
C. 2.291%
D. 1.876%

75. The current stock price of LLP Limited is $22. The volatility is 20%, and the risk-free rate is 7%. According to the Black-Scholes-Merton model, what should be the price of a European call option on LLP's stocks with a strike price of $20 and expiration of 3 months if $N(d_1)$ and $N(d_2)$ are assumed to be 0.5522 and 0.5027, respectively?

A. $2.27
B. $2.59
C. $2.98
D. $4.33

76. Five years ago, an investment bank provided a $100,000,000 loan to RMS Constructions. The loan has the floating rate of 5% + 1-year Libor. The interest is paid annually, and the last interest payment was paid on the last day of 2016. Below is the information available for the 1-year Libor rates:

31/12/15	0.5%
31/12/16	2.1%
31/12/17 (forecast)	3.3%

What is the accrued interest as of 16/07/2017 if the bank and the client agreed to use the ACTUAL/360 day count convention?

A. $3,009,722.2
B. $3,885,277.8
C. $4,268,333.3
D. $4,541,944.4

77. A fixed-income trader summarizes in the table below the prices of Treasury Bonds with semiannual coupon payment. The data is as of 01/01/17.

	Maturity	Coupon Rate	Price (per 100 face value)
Tranche 1	30/06/2017	3.5%	99-00
Tranche 2	31/12/2017	4%	100-16
Tranche 3	30/06/2018	5%	101-04

What are the discount factors for 0.5, 1, and 1.5 years?

A. d(0.5) = 0.9730; d(1) = 0.9662; d(1.5) = 0.9393
B. d(0.5) = 0.9551; d(1) = 0.9422; d(1.5) = 0.9102
C. d(0.5) = 0.9633; d(1) = 0.9523; d(1.5) = 0.9085
D. d(0.5) = 0.98990; d(1) = 0.9782; d(1.5) = 0.8787

78. An investment bank has a position in a callable bond. The risk management team of the bank prepared three different scenarios. The results are summarized in the table below:

Interest rate	Price of the callable bond	Price of the call option
8.70%	$101.750	$1.19
9.00%	$98.710	$1.01
9.30%	$95.988	$0.85

The CRO of the bank is evaluating the proposal of the structuring team to bifurcate the call option (e.g., to separate the call option from the bond and treat it as two different products: a non-callable bond and a call option) and sell it on the market. The CRO is primarily concerned with the impact of the bifurcation on the duration of the bond.

What is the effective duration (D) of the callable bond and a comparable non-callable bond?

A. D(callable bond) = 9.4384; D(non-callable bond) = 10.1986
B. D(callable bond) = 9.7288; D(non-callable bond) = 10.1986
C. D(callable bond) = 9.7288; D(non-callable bond) = 11.2873
D. D(callable bond) = 9.4384; D(non-callable bond) = 11.2873

79. A trader uses a model uses his valuation model to estimate the value of a bond portfolio at CAD 276.060 million. The term structure is flat. Using the same model, he estimates that the value of the portfolio would increase to CAD 278.935 million if all interest rates fell by 40 bps and would decrease to CAD 272.650 million if all interest rates rose by 40 bps. Using these estimates, determine the effective duration of the bond.

A. 2.85
B. 0.028
C. 0.023
D. 0.285

80. John Wick, FRM, manages two assets – A and B. The correlation between A and B is 0.2. The table below outlines further details on the two assets:

Asset	Expected Return	Annual Volatility	Value
A	20%	20%	$200
B	30%	25%	$100

If Wick sells $100 worth of A and buys $100 worth of B, what would be the change in the daily VaR at the 99% level of confidence? Assume a normal distribution and 260 trading days.

A. 8.000
B. 0.897
C. 7.403
D. 0.102

81. A trader wants to price a bond with a 5% coupon rate and a maturity of 1.5 years. The coupon payment is semiannual. Due to a bug in the system, the trader was not able to download the complete structure of the market spot rates.

The table below shows the data the trader was able to gather.

Term in years	Spot rates
0.5	1.10%
1.0	n/a
1.5	3.00%

Which of the following are the missing spot rate and the bond price if the 1-year discount factor is 0.9831?

A. 1-year spot rate = 1.71%; Bond price = 102.967
B. 1-year spot rate = 2.10%; Bond price = 102.957
C. 1-year spot rate = 2.50%; Bond price = 102.947
D. 1-year spot rate = 2.65%; Bond price = 104.944

82. An analyst at a bond trading desk wants to decompose the P&L of a $5,000,000 face value U.S. Treasury 7 1/2s of April 31, 2030, over a six-month holding period. Currently, the bond trades at par. The position is financed with direct repo which has an interest rate of 3%.

If the market experiences a +100 basis-point parallel shift in the yield curve at the end of the sixth month, what is the cash-carry amount over the six month period?

A. -$500,000
B. -$75,000
C. +$112,500
D. +$187,500

83. Amelia Stone plans to retire next year and is considering an investment that would help to ensure her future financial stability. On average, Stone spends $70,000 annually. After retirement, she plans to relocate to the countryside, a move that should cut her living expenses by 20%. She was approached by her broker with the idea to start investing in perpetuities. Specifically, he suggested a perpetuity that would yield 7% per year.

Assuming the mentioned reduction of costs, how much should Stone invest in the perpetuities to meet her new spending needs?

A. $800,000
B. $8,000,000
C. $560,000
D. $5,600,000

84. A risk manager in an investment bank was asked to forecast the sovereign rating of countries where the bank has significant holdings. The risk manager found sovereign transition rates across the Major Rating Categories 1995-2008 published by Fitch.

Annual Rating Transitions (%, Average Annual)

	AAA	AA	A	BBB	BB	B	CCC to C	D	Total
AAA	99.42	0.58	-	-	-	-	-	-	100.00
AA	4.12	94.12	1.18	-	-	0.59	-	-	100.00
A	-	3.55	92.91	3.55	-	-	-	-	100.00
BBB	-	-	8.11	87.84	3.38	0.68	-	-	100.00
BB	-	-	-	9.04	83.51	5.85	-	1.60	100.00
B	-	-	-	-	12.12	84.09	3.03	0.76	100.00
CCC to C	-	-	-	-	-	23.08	53.85	23.08	100.00

Given the table above, which of the below statements is correct?

A. A country has a 98.13% chance to have a rating of AA.
B. A country with a rating of BB has a 5.85% chance of a downgrade in one year.
C. AAA-rated countries will have the same rating in the future with a 99.42% probability.
D. There is a 7.97% probability that a country with a rating of AA will have an improvement of rating to AAA in two years.

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85. For which of the following reasons are rating agencies most usually criticized?

A. Being too optimistic in the assessment of corporate ratings and too pessimistic in the assessment of sovereign ratings
B. Failure to align ratings with each other before publishing (e.g., one rating agency may downgrade a sovereign rating of the country to speculate grade, while the other can still keep it with investment grade)
C. Taking too long to change a rating
D. Charging high fees for publishing corporate and sovereign ratings

86. A bank has the following internal rating transition matrix:

Annual Rating Transitions (%, Average Annual)

	A	**B**	**C**	**D**
A	98.50	1.00	0.50	-
B	1.00	88.00	7.50	3.50
C	-	10.00	75.50	14.50

Based on the matrix, what is the probability of default over 2 years of a company with a rating of B?

A. 2.5%.
B. 4.6%.
C. 6.8%.
D. 7.7%.

87. An intern in a risk management department of a bank is struggling with the understanding of the nature of credit ratings provided by the rating agencies. He overhears a discussion from some of his colleagues regarding the recent downgrade of company XYZ by S&P from A to BBB.

What is the interpretation of XYZ's credit rating downgrade from A to BBB?

A. Bonds issued by XYZ are currently overvalued in the market
B. Stocks of XYZ are expected to underperform the market benchmark in the nearest future
C. XYZ's capacity to meet its obligations slightly deteriorated from 'extremely strong' to 'very strong'
D. XYZ still has an adequate capacity to meet its obligations, but now it is more susceptible to a negative change in the business environment

88. What is the role of the internal audit in stress testing governance?

A. Establishing adequate stress testing policies and procedures and ensure compliance with them
B. Monthly statistical back-testing of the stress test estimates against realized outcomes
C. Provision of an independent evaluation of the ongoing performance, integrity, and reliability of stress-testing activities
D. Appropriate resource allocation to ensure proper functioning of stress-test processes

89. The new CRO of a regional bank is trying to implement a new stress testing process for the bank. He asks one of the colleagues to start the preparation of stress testing policies, procedures and documentations. The CRO makes the following statements:

I. The policies should be transparent to third parties to ensure an understanding of the bank's stress testing activities.
II. The policies should describe the frequency and priority with which stress-testing activities should be conducted.
III. The policies should be reviewed and updated as necessary to ensure that stress testing practices remain up to date with changes in market conditions and the bank's strategies.
IV. The policies should describe the overall purpose of stress-testing activities.

Which of the statements is/are correct?

A. Statement I and
B. Statement II and III
C. Statement II, III and IV
D. All of the statements are correct

90. An investment bank has $1,000,000,000 portfolio of senior unsecured loans evenly distributed between 10 different clients, each with an internal rating B. The CRO of the bank wants to stress the loan portfolio under a scenario of an economic slowdown with the GDP dropping by 2% and the unemployment increasing by 1%. Based on the prior experience, the CRO knows the economic slowdown scenario will negatively impact the credit quality of the borrowers, and lead to a rating deterioration from B to C for half of the clients. The recovery rates of senior unsecured facilities are assumed to be 60% in normal conditions and 40% in stress conditions.

Credit Rating	**PD**
A	0%
B	3%
C	25%
D	100%

What is the bank's loan portfolio expected loss in both the base and the stressed scenarios?

A. Loss (*Base scenario*) = $12,000,000; Loss (*Stressed scenario*) = $56,000,000
B. Loss (*Base scenario*) = $12,000,000; Loss (*Stressed scenario*) = $84,000,000.
C. Loss (*Base scenario*) = $18,000,000; Loss (*Stressed scenario*) = $56,000,000.
D. Loss (*Base scenario*) = $18,000,000; Loss (*Stressed scenario*) = $84,000,000.

91. You have been provided the following information:

Weighted long-run variance	0.000010
Weighting on previous period's return	0.15
Weighting on previous volatility estimate	0.80
Daily volatility estimate	1.5%
Recent market return	1%

Using the GARCH(1,1) model, the volatility per day on day n (σ_n) is closest to:

A. 1.109%
B. 1.276%
C. 1.432%
D. 1.231%

92. Risk managers of a regional bank are discussing the advantages of stress testing. They make the following statements:

I. Based solely on historical data, stress testing provides forward-looking assessments of risk.
II. Stress testing plays an important role in capital and liquidity planning procedures.
III. Stress testing is facilitating the development of risk mitigation or contingency plans across a range of stressed conditions

Which of these statements is/are correct?

A. Statement I.
B. Statements I and II.
C. Statements I and III.
D. Statements II and III.

93. Which of the following statements regarding recommendations to supervisors is incorrect?

A. Supervisors should evaluate whether the scenarios are consistent with the risk appetite the bank has set for itself
B. Supervisors should take corrective actions if material deficiencies in the stress testing program are identified or if the results of stress tests are not adequately taken into consideration in the decision-making process
C. Supervisors should verify the active involvement of senior management in the stress testing program and require a bank to submit at regular intervals the results of its firm-wide stress testing program
D. Supervisors should verify that stress testing forms an integral part of the ICAAP and the bank's liquidity risk management framework

94. In the run-up to the 2007/2008 financial crisis, Trevor Smith inherited $10,000,000 from his grandfather. During the crisis, Smith suffered huge losses in his equity holdings, and he now wants to invest the inheritance with minimal risks. He is considering the following options:

	Product	Interest rate	Interest rate compounding
Option 1	Deposit at an AA-rated bank	3.0500%	monthly
Option 2	Deposit at an AA-rated bank	3.1000%	semiannually
Option 3	Deposit at an AA-rated bank	3.1000%	annually
Option 4	Deposit at an AAA-rated bank	3.0500%	continuously

Which of these options has the highest monthly compounded interest rate?

A. Option 1 has the highest monthly compounded rate of 3.0500%.
B. Option 2 has the highest monthly compounded rate of 3.0802%.
C. Option 3 has the highest monthly compounded rate of 3.0868%.
D. Option 4 has the highest monthly compounded rate of 3.0539%.

95. The stock price of AlphaFarm skyrocketed to $200 after a recent announcement of a drug approval by the FDA. Francesca Merc wants to quickly estimate the approximate price of her European call option on Alpha's stocks using the Black-Scholes-Merton model. The call option has expiry in 3 months and a strike price of $50.

If the risk-free rate is 8%, then which of the following is closest to the price of Merc's call option?

A. $151
B. $158
C. $147
D. $140

96. If the covariance between Canadian and American interest rates is 0.092, and the variances of interest rates in Canada and the U.S. are 12.32% and 11.04%, respectively, then which of the following is closest to the correlation between Canadian and American interest rates?

A. 0.7889
B. 9.7641
C. 0.1478
D. 1.2677

97. Consider a \$75 call option on a non-dividend paying stock where the stock price is \$68.30, the risk-free rate is 5%, the time to maturity is 120 days and N' = 0.3311. A 1% increase in the volatility will increase the value of the option by approximately:

A. 0.1424
B. 0.0743
C. 0.1297
D. 2.4873

98. Arab Bank has taken credit exposure to two corporate clients. The credit risk characteristics of these two loans have been provided to you:

Loan to BMG Electrics

Sanctioned amount	USD 730 million
Exposure amount	USD 630 million
Probability of default over the next one year	2.1%
Loss rate if the customer defaults	24%
Standard deviation of the probability of default	5%
Standard deviation of the loss rate	38%

Loan to FFA Construction

Sanctioned amount	USD 525 million
Exposure amount	USD 300 million
Probability of default over the next one year	1%
Loss rate if the customer defaults	25%
Standard deviation of the probability of default	3%
Standard deviation of the loss rate	25%

The correlation between the two loan accounts is 0.45.

What is the unexpected loss of the loan portfolio held by Nicolson Finance?

A. USD 48.86 million
B. USD 38.03 million
C. USD 39.65 million
D. USD 17.13 million

99. Bakster Bank is following the basic indicator approach for calculating the operational risk amount for the year 2016. Some past financial details of the bank are given below:

Year	Net interest income	Non-interest income
2015	102	25
2014	104	22
2013	98	21
2012	91	17

(All $ amounts are in USD millions)

Based on the original Basel Accord, the bank must hold capital for operational risk for 2016 equal to:

A. USD 18 million
B. USD 18.6 million
C. USD 24.8 million
D. USD 31 million

100. Elisabeth Yung is the senior fixed income investment manager at a regional bank. She is interested in the investment in Rocks Corporation's medium-term (7-year) zero-coupon bond. The bond has a yield to maturity of 4.5% compounded annually and a daily yield volatility of 0.59%. What are the daily 99% VaR and the Macaulay duration of the bond if Yung decides to buy the nominal value of $10,000,000?

A. Daily 99% VaR = $920,851.67; Macaulay duration = 7
B. Daily 99% VaR = $676,668.02; Macaulay duration = 7
C. Daily 99% VaR = $676,668.02; Macaulay duration = 6.7
D. Daily 99% VaR = $920,851.67; Macaulay duration = 6.7